

Register-Closure π : Emergent Discretization from Computational Substrate

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September 2025

1 Introduction

The computational substrate requires no transcendental functions— π emerges from pure register algebra through closure conditions. This demonstrates that π , like h and α , is not an input constant but an *architectural property* arising from the discrete phase ladder of the six-register system.

2 Rational Rotation Construction

Given parameter $t = \tan(\theta/2)$, define rotation via purely algebraic operations (no trigonometry):

$$c = \frac{1 - t^2}{1 + t^2}, \quad s = \frac{2t}{1 + t^2} \quad (1)$$

These represent $(\cos \theta, \sin \theta)$ computed from the rational parameter t using only arithmetic operations.

3 Composition Through Exponentiation

Compose N rotations through fast exponentiation (no trigonometry required):

$$(c, s)^N = \underbrace{(c, s) \otimes (c, s) \otimes \cdots \otimes (c, s)}_{N \text{ times}} \quad (2)$$

where the composition operator is defined algebraically:

$$(c_1, s_1) \otimes (c_2, s_2) = (c_1 c_2 - s_1 s_2, s_1 c_2 + c_1 s_2) \quad (3)$$

This is rotation multiplication in (c, s) algebra, computed purely through rational arithmetic on the parameter t .

4 Closure Condition

Find the smallest $t > 0$ such that:

$$s_N = 0 \tag{4}$$

$$c_N > 0 \tag{5}$$

This corresponds to $N\theta = 2\pi$ (full rotation returning to identity) *without invoking π itself*. The condition is detected through sign changes in s_N during a rational scan over t .

5 Extracting π from the Closure

Once the closure parameter t is found, extract π from the inscribed polygon perimeter:

$$\pi_{\text{eff}}(N) = N \cdot \frac{2t}{\sqrt{1+t^2}} \tag{6}$$

The chord length $\frac{2t}{\sqrt{1+t^2}} = 2\sin(\theta/2)$ is computed using only the rational parameter t and a square root—no trigonometric functions.

As $N \rightarrow \infty$, $\pi_{\text{eff}}(N) \rightarrow \pi$ through purely rational operations.

6 Physical Interpretation

The phase register Φ in the six-register substrate has discrete quantization levels determined by capacity constraints (10^6 packets per Planck tick). Rotations accumulate in Φ until closure—the register returns to its initial state.

The effective π_{eff} observed at finite N reflects the *discrete phase ladder* of the substrate:

- At small N (coarse quantization): $\pi_{\text{eff}}(N)$ deviates from continuous π
- At large N (fine quantization): $\pi_{\text{eff}}(N) \rightarrow \pi$ asymptotically
- True continuous π is the infinite-capacity limit
- Physical systems operate at finite N , yielding observable discretization in rotational symmetry

7 Connection to Codec Operations

This discrete rotational algebra connects directly to codec operations:

- **Codec M_7 (Phase Rotation):** Implements (c, s) composition on the Φ register. The phase increment per tick is quantized by the closure period.
- **Codec M_9 (Entanglement):** Phase synchronization between spatially separated registers relies on matching discrete phase values modulo the closure period.
- **Fermion 720° cycle:** The $SU(2)$ double cover (spinor behavior) emerges from the discrete phase ladder. A fermion requires two full rotations (4π in phase) to return to identity in the register algebra.
- **Pauli exclusion:** Hash collision detection in (Φ, X, C) space depends on discrete phase quantization. Two fermions cannot occupy the same discrete phase state.

Angular momentum quantization and fermionic statistics emerge from this discrete rotational algebra, not from postulated quantum rules.

8 Comparison with Classical Methods

Traditional π computation (Chudnovsky, Ramanujan, etc.) treats π as a transcendental constant to be approximated. The register-closure method reveals π as:

$$\pi = \lim_{N \rightarrow \infty} \pi_{\text{eff}}(N) = \lim_{N \rightarrow \infty} \left[N \cdot \frac{2t_N}{\sqrt{1 + t_N^2}} \right] \quad (7)$$

where t_N is found by solving the purely algebraic closure condition $(c, s)^N = (1, 0)$ with $c > 0$.

No transcendental functions are used in the computation—only:

- Rational arithmetic on parameter t
- Square root (algebraic, not transcendental)
- Integer exponentiation via repeated composition
- Sign-change detection (bisection)

9 Key Result

π is not an input constant—it emerges as the *period of register composition* under the closure constraint. Like h and α , it is **architectural**.

The computational substrate does not require π to be specified externally. Instead, π appears naturally as:

- The closure period of phase register rotation
- The ratio of polygon perimeter to diameter in the $N \rightarrow \infty$ limit
- A consequence of discrete quantization in the Φ register

Just as h emerges from the 10^6 packet-per-tick capacity constraint, and α from the scalar-to-vector cost ratio, π emerges from the discrete phase ladder imposed by finite register capacity.

10 Testable Predictions

If physical reality operates on a discrete computational substrate with finite capacity, rotational symmetry should exhibit subtle discretization:

1. **Effective π in high-energy scattering:** At energies approaching Planck scale, measured angular distributions may show deviations from continuous rotational symmetry, corresponding to $\pi_{\text{eff}}(N)$ for some finite N .
2. **Phase quantization in interferometry:** Ultra-precision phase measurements might detect discrete jumps corresponding to the phase ladder, rather than perfectly continuous rotation.
3. **Anomalous angular momentum:** Measured spin values at extreme precision could show slight deviations from integer/half-integer multiples of $\hbar$ if the underlying phase register has finite resolution.

These effects would be suppressed at macroscopic scales (large N) but could appear near the computational limits of the substrate.

11 Conclusion

The register-closure method demonstrates that π is not a fundamental input to physics but an *emergent property* of the discrete phase register algebra. By computing π through purely rational operations on closure conditions, we reveal its status as an architectural constant alongside h and α .

Critical insight: In the discrete computational substrate, $\pi_{\text{eff}}(N)$ is *not irrational*. For any finite closure order N :

$$\pi_{\text{eff}}(N) = N \cdot \frac{2t_N}{\sqrt{1 + t_N^2}} \quad (8)$$

where t_N is a *rational or algebraic number* satisfying the closure condition. Since t_N is found through rational bisection and the formula involves only multiplication and one square root, $\pi_{\text{eff}}(N)$ is an **algebraic number**, not a transcendental.

The apparent irrationality of π arises only in the infinite-capacity limit $N \rightarrow \infty$. At finite N (which is all that physical reality can implement), the effective π is:

- **Computable** from finite rational operations
- **Algebraic** (root of a polynomial equation)
- **Approximate** to the transcendental π , with error $\sim 1/N^2$

The substrate does not contain transcendental π —it contains $\pi_{\text{eff}}(N)$ for $N \sim 10^6$ (packets per tick). This is a *rational approximation* that is indistinguishable from π at accessible precision, but fundamentally different in character.

Just as spacetime is discrete at Planck scale, **rotational symmetry is discrete** at the register level. There is no continuous circle—only a polygon with N sides. The "circle" we observe is the large- N limit of this discrete structure.

Philosophical implication: Transcendental numbers like π and e do not exist in physical reality. They are mathematical idealizations of the $N \rightarrow \infty$ limit. The universe computes with algebraic numbers only, because it has finite capacity.

*The universe computes rotations rationally.
 π is what emerges when you close the loop.
 But at finite N , that π is algebraic, not transcendental.*